Econometrics Chapter 7: Hypothesis Testing

Note: Y_i, β_i and ε_i same distribution but centered about diff mean

(1) Hypothesis Testing in MLR

Single Hypothesis

- to test $H_0: \beta_1 = 0$ vs $H_A: \beta_1 \neq 0$

  $H_1: \beta_2 = 0$ vs $H_A: \beta_2 \neq 0$

  $H_2: \beta_1 + \beta_2 = 1$ vs $H_A: \beta_1 + \beta_2 \neq 1$

  [constant return to scale]

Joint Hypothesis

- to test: $H_0: \beta_1 = 0$ and $\beta_2 = 0$

  $H_A: \beta_1 \neq 0$ or $\beta_2 \neq 0$ (or both)

Assuming Multivariate Normal Error

- $E(\varepsilon) = 0$ $\quad Var(\varepsilon) = \Omega$ $\quad \varepsilon \sim N(0, \Omega)$
  - homo — $\Omega = \text{trace}[\sigma^2 \dots \sigma^2]$
  - hetero — $\Omega = \text{trace}[\sigma_1^2 \dots \sigma_n^2]$
- given pdf of normal error

$$f(\varepsilon) = (2\pi)^{-\frac{n}{2}} |\Omega|^{-\frac{1}{2}} \exp\left[-\tfrac{1}{2}(\varepsilon-\mu)^T \Omega^{-1} (\varepsilon-\mu)\right]$$

(determinant)

- applying assumption $E(\varepsilon) = 0$

$$f(\varepsilon) = (2\pi)^{-\frac{n}{2}} (\det \Omega)^{-\frac{1}{2}} \exp\left[-\tfrac{1}{2}\varepsilon^T \Omega^{-1} \varepsilon\right]$$

- applying assumption $E(\varepsilon_i \varepsilon_j) = 0$
  - $|\Omega| = \prod_{i=1}^{n} \sigma_i^2$
  - $\Omega^{-1} = \text{diag}(1/\sigma_1^2 \dots 1/\sigma_n^2)$
  - $\varepsilon^T \Omega^{-1} \varepsilon = \sum_{i=1}^{n} \frac{\varepsilon_i^2}{\sigma_i^2}$

$$f(\varepsilon) = (2\pi)\left(\prod_{i=1}^{n} \sigma_i^2\right)^{-\frac{1}{2}} \exp\left(-\tfrac{1}{2}\sum_{i=1}^{n} \frac{\varepsilon_i^2}{\sigma_i^2}\right)$$

- Homoskedastic Error $\sigma_i^2 = \sigma^2$

$$f(\varepsilon) = (2\pi)\left(\prod_{i=1}^{n} \sigma^2\right)^{-\frac{n}{2}} \exp\left(-\frac{1}{2\sigma^2}\sum \varepsilon_i^2\right)$$

∴ Observe each ε is a function of itself.

Conclude that ε_i is not related to ε_j, errors can be assume to be normally distributed multivariately.

Case 1: Hypothesis Testing with Normal Errors

- assume $\varepsilon | X \sim N(0, \sigma^2 I)$

a. General single linear hypothesis

$H_0: r^T\beta = r_0$ vs $H_A: r^T\beta \neq r_0$

- to test $\beta_1 = 0$, set $r^T = [0 \mid 0]$ and $r_0 = 0$
- to test $\beta_2 = 1$, set $r^T = [0\ 0\ 1]$ and $r_0 = 1$ [Proportionate Rs]
- to test $\beta_1 + \beta_2 = 1$, set $r^T = [0\ 1\ 1]$ and $r_0 = 1$ [CRS]
- to test $\beta_1 = \beta_2$, set $r^T = [0\ 1\ -1]$ and $r_0 = 0$

- Since ε is distributed Multivariate Normal

  $a_1\varepsilon_1 + a_2\varepsilon_2 + \ldots + a_n\varepsilon_n$ remains normally distributed

$\hat{\beta}^{ols} = \beta + (X^TX)^{-1}X^T\varepsilon$

$\hat{\beta}^{ols} | X$ can be assume to be Normal $\sim N_k(\beta, \sigma^2(X^TX)^{-1})$

- $N_k(\beta, \sigma^2(X^TX)^{-1})$ could be used to develop t-test & F-test

- Expansion:

$\hat{\beta}^{ols} | X \sim N_k(\beta, \sigma^2(X^TX)^{-1})$

$r^T\hat{\beta}^{ols} | X \sim N_k(r^T\beta, r^T(\sigma^2(X^TX)^{-1})r)$

- Null Hypothesis (If it holds)

$H_0: r^T\beta = r_0$ holds

$r^T\hat{\beta}^{ols} | X \sim N(r_0, r^T(\sigma^2(X^TX)^{-1})r)$

Z statistics: $\dfrac{r^T\hat{\beta} - r_0}{\sqrt{r^T(\sigma^2(X^TX)^{-1})r}} \sim N(0,1)$ σ^2 known

t statistics: $\dfrac{r^T\hat{\beta} - r_0}{\sqrt{r^T(\hat{\sigma}^2(X^TX)^{-1})r}} \sim t_{(n-k-1)}$ σ^2 unknown

b. Multiple Hypotheses [F-test]

$H_0: R\beta = r_0$ vs $H_A: R\beta \neq r_0$

- whereby, R is a $J \times k$ matrix and r_0 is a $J \times 1$ vector

- Example: $H_0: \beta_1 + \beta_2 = 1$ and $\beta_3 = 0$

$H_A: \beta_1 + \beta_2 \neq 1$ and $\beta_3 \neq 0$

In this case: Set $R = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$r_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

- F-test: Compares SSR from the restricted and unrestricted regression
  - note: Restricted SSR will never be smaller than unres SSR

F statistic: $\dfrac{(SSR_{res} - SSR_{unres})/J}{SSR_{unres}/n-k-1}$

$$= \frac{(\hat{\varepsilon}_{res}^T \hat{\varepsilon}_{res} - \hat{\varepsilon}_{ols}^T \hat{\varepsilon}_{ols}}{\hat{\varepsilon}_{ols}^T \hat{\varepsilon}_{ols}/n-k-1} \sim F(J, n-k-1)$$

- J is the no. of restrictions

Alt Expression:

$$F = \frac{(R^2 - R^2_{res})/J}{(1-R^2)/(n-k-1)}$$

- Full Example:

Model: $Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i3} + \varepsilon_i$

Hypotheses: $H_0: \beta_1 + \beta_2 = 1 \quad \beta_3 = 0$

Step 1: Estimate unrestricted regression

$\hat{\varepsilon}^{ols\,T} \hat{\varepsilon}^{ols} = SSR_{unres}$

Step 2: Estimate Restricted Regression

Force $\beta_3 = 0$ and $\beta_1 = 1 - \beta_2$

$Y_i = \beta_0 + (1-\beta_2) X_{i1} + \beta_2 X_{i2} + 0 + \varepsilon_i$

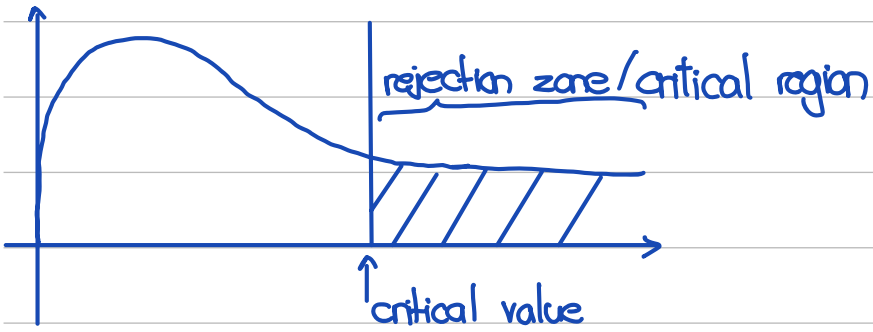
$\hat{Y}_{ires} = \hat{\beta}_0 + (1-\hat{\beta}_2) X_{i1} + \hat{\beta}_2 X_{i2}$

$\hat{\varepsilon}_{ires} = Y_i - \hat{Y}_{ires}$

$SSR_{res} = \hat{\varepsilon}_{ires}^T \hat{\varepsilon}_{ires}$ (increases, but if restriction does not hold it increases by alot)

$\therefore J = 2, \; k = 4$

Rejection zone / Critical Region



F-stats

- $\hat{\varepsilon}_{res}^T \hat{\varepsilon}_{res} - \hat{\varepsilon}_{ols}^T \hat{\varepsilon}_{ols} = (R\hat{\beta} - r_0)^T [R(X^TX)^{-1}R]^{-1}(R\hat{\beta}^{ols} - r_0)$
- $F_{test} = (R\hat{\beta} - r_0)^T [R\hat{\sigma}^2 (X^TX)^{-1} R^T]^{-1} (R\hat{\beta}^{ols} - r_0)/J$

  $= (R\hat{\beta} - r_0)^T [R\,Var(\hat{\beta}^{ols}|X)R^T]^{-1}(R\hat{\beta}^{ols} - r_0)/J \sim F_{(J, n-k-1)}$
- $F\text{-test} = (t\text{-test})^2$

Case 2: Others — non-normality or Heteroskedasticity
Do not assume $\varepsilon|X$ to be Normal

a. Non normal cases

- appeal to CLT
  - t statistic is approximately normal
  - $JF \approx W \sim \chi^2_J$ when n is large [Wald Stats: Squared dist of estimate from res.]

  when $J=1$, we can use t

  $J>1$, use chi-square

b. Heteroskedasticity

- utilize het robust variances to construct het robust t and F statistics

  Replace $\widehat{Var}(\hat{\beta}^{ols}|X)$ with het robust variance estimator in t and F statistics

(2) Specification Tests

a. Testing for Heteroskedasticity

(Test 1) Breusch-Pagan Test

- test if σ_i^2 depends on regressor
  use $\hat{\varepsilon}_{i,ols}^2$ as proxy for $Var(\varepsilon_i)$

Step 1: Estimate main regression by OLS and obtain OLS residuals $\hat{\varepsilon}_{i\,ols}$

Step 2: Breusch-Pagan Test

$$\hat{\varepsilon}_{i,ols}^2 = \alpha_0 + \alpha_{11} X_{i1} + \dots + \alpha_{1k} X_{ik-1} + u_i$$

and test if $H_0: \alpha_0 = \dots = \alpha_{k-1} = 0$ using F-test

Step 3: LM test (Lagrangian Multiplier test)

$LM = nR^2_{aux}$ — when sample is large under CLT

whereby, aux regression is $\hat{\varepsilon}_{i,ols}^2 = \alpha_0 + \alpha_{11} X_{i1} + \dots + \alpha_{k-1} X_{ik-1} + u_i$

R^2_{aux} is taken from the aux regression estimate by OLS

Step 4: Chi-Square Distribution Comparison

$$nR^2_{aux} \xrightarrow{d} \chi^2_{k-1}$$

$p\text{-value} = P(\chi^2_{k-1} > LM)$ — p-value is the probability that a chi-square random variable is greater than your observed LM value

Reject H_0 if $p < 0.05$

reject H_0 if your observed LM is larger than the 95th percentile of the chi-square distribution

Non-linear Forms

$$\hat{\varepsilon}_i^2 \,ols = \alpha_0 + \alpha_1 X_{i1} + \dots + \alpha_{k-1} X_{k-1}$$
$$+ \delta_1 X_{i1}^2 + \dots + \delta_{k-1} X_{ik-1}^2$$
$$+ \gamma_{12} X_{i1} X_{i2} + \dots + \gamma_{1k} X_{i1} X_{ik-1}$$
$$+ \dots + u_i$$

- test if all slope coefficients are 0

∴ Become infeasible pretty quickly

(Test 2): White Test

- more feasible than Breusch-Pagan test

Step 1: Estimate main regression by OLS and obtain $\hat{\varepsilon}_{i,ols}$ and $\hat{Y}_{i,ols}$

Step 2: White test

Regress $\hat{\varepsilon}_{i\,ols}$ on $\hat{Y}_{i\,ols}$ — doesn't pinpoint which way $\hat{\varepsilon}_{i\,ols}$ is related with X

Simplest Form: $\alpha_0 + \alpha_1 \hat{Y}_{i\,ols} + \alpha_2 \hat{Y}_{i\,ols} + u_i$

Step 3: LM test

$LM = nR_{\varepsilon}^2$

- n is the no. of observation
- $R\varepsilon^2$ is the R^2 of the regression of squared residuals on the dependent variables

Step 4: Chi-Square Distribution Comparison

$nR^2_{aux} \xrightarrow{d} \chi^2_{k-1}$

p-value $= P(\chi^2_{k-1} > LM)$

Reject H_0 if $p < 0.05$

Or Alternatively,

Step 3: Test $H_0: \alpha_1 = \alpha_2 = 0$ using F-test

For Both Tests:

- p-value is low, null will be rejected (error is heteroskedastic)
- p-value is high, null will not be rejected (error is not heteroskedastic)

b. Testing for Functional Form Misspecification

Test: RESET TEST

- linear-in-regressor specification may be incorrect

$$Y_i = \beta_0 + \beta_1 X_{i1} + \ldots + \beta_k X_{ik} + \varepsilon_i$$

- RESET Test attempts to involve adding powers of the fitted values or other transformation of the independent variables to the orginal regression model see if these additional terms are statistically significant

Should additional terms be statistically significant, it suggest that the original specification may be inadequate and model is misspecified

- Test eq^n: $Y_i = \beta_0 + \beta_1 X_{i1} + \ldots + \beta_k X_{ik} + \alpha_2 \hat{Y}_{i\,ols}^2 + \ldots + \alpha_p \hat{Y}_{i\,ols}^p$ — unres model

Note: $\hat{Y}_{i\,ols}$ is not included in the auxiliary regression

Step 1: Run initial regression

Obtain Fitted Values from initial regression

Step 2: Run auxiliary regression

Step 3: Test for significance using an F-test and see if additional terms are significant

$H_0: \alpha_2 = \ldots = \alpha_p = 0$

$$F = \frac{(SSR_r - SSR_u)/J}{SSR_u/(n-k-J-1)}$$

whereby, J is the no. of non-linear terms added

c. Testing for Non-nested Alternatives

Non-nested Alternative Test

- comparing two distinct models that are not subsets of each other (Neither model is a restricted ver of the other)
  Assumes one model's variable or functional form is correct

- Models:

[A] $Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \varepsilon_i$

[B] $Y_i = \beta_0 + \beta_1 \ln X_{i1} + \beta_2 \ln X_{i2} + \varepsilon_i$

[Test] Davidson-MacKinnon J-test (Testing explanatory power of each model)

- step 1: Run OLS Regression of both models and collect fitted value $\hat{Y}_{iA}$ $\hat{Y}_{iB}$

Step 2: Fit both model by including other fitted values as their regressor

Step 3: New Model:

[A']: $Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \delta_1 \hat{Y}_{i,B} + \varepsilon_i$

[B']: $Y_i = \beta_0 + \beta_1 \ln X_{i1} + \beta_2 \ln X_{i2} + \delta_2 \hat{Y}_{i,A} + \varepsilon_i$

Step 4: Test Seperately if fitted values are stastically significant using F-test

Note:
- Both insignificant: both fit model well
- Both significant: Each contains information not captured by the other
  Highlighting the need for an alternative model

d. Testing Normality of Noise Terms

Setup

If $X \sim N(\mu, \sigma^2)$, then

Skewness $S = E((X-\mu)^3)/\sigma^3 = 0$

Kurtosis $K = E((X-\mu)^4)/\sigma^4 = 3$ $t > 3$ [Fat tails]

Estimation is done by

$$\hat{S} = \frac{\frac{1}{N}\sum(x_i - \bar{x})^3}{\left[\frac{1}{N}\sum(x_i - \bar{x})^2\right]^{3/2}}$$

$$\hat{K} = \frac{\frac{1}{N}\sum(x_i - \bar{x})^4}{\left[\frac{1}{N}\sum(x_i - \bar{x})^2\right]^2}$$

Test: Jarque-Bera Test — testing if ε is normal

- Step 1: Form JB

$$JB = \frac{n-k-1}{6}\left(\hat{S}^2 + \frac{1}{4}(\hat{K}-3)^2\right) \xrightarrow{d} \chi^2_{(2)}$$

Step 2: p-value

p-value $= P(\chi^2_2 > JB)$

- small p, reject null, ε is not normal
- large p, accept null, ε is normal